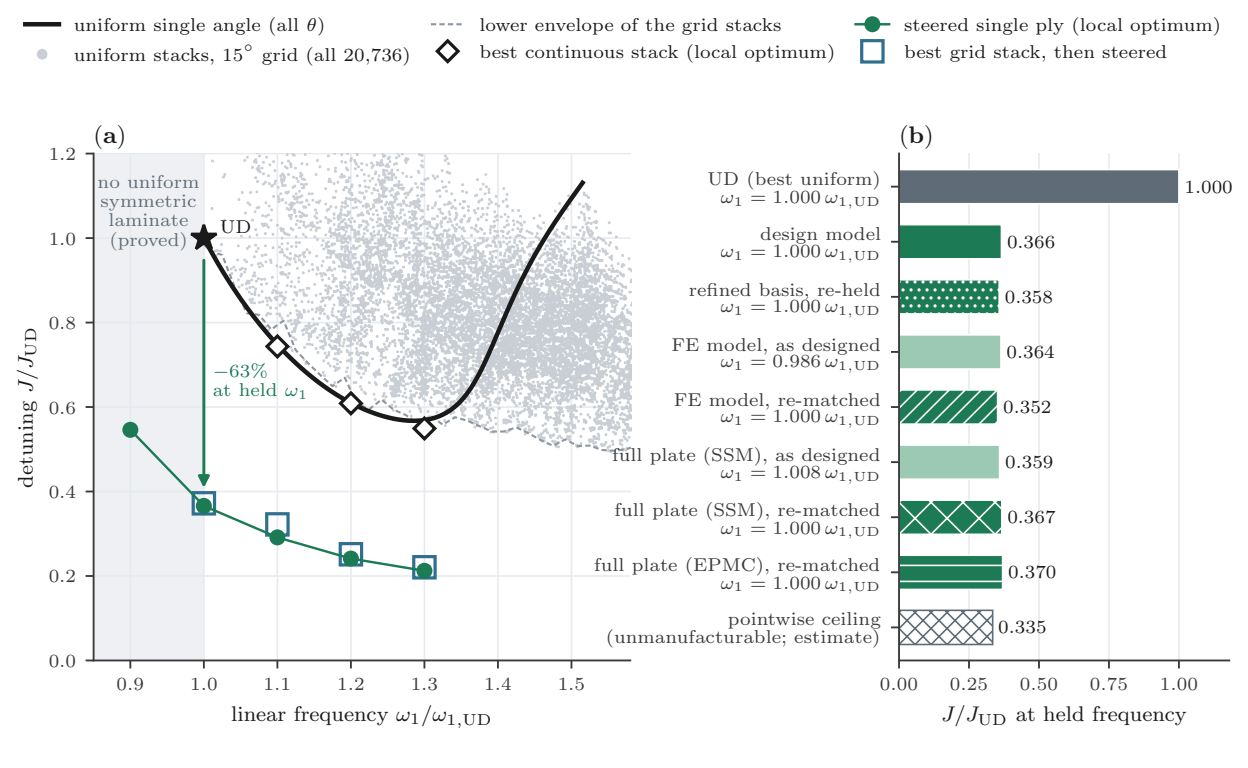

uniform single angle (all θ)
uniform stacks, 15° grid (all 20,736)
lower envelope of the grid stacks
best continuous stack (local optimum)
steered single ply (local optimum)
best grid stack, then steered
(a)
no uniform symmetric laminate (proved)
UD
−63% at held ω1
detuning J/J_UD
linear frequency ω1/ω1,UD
(b)
UD (best uniform) ω1 = 1.000 ω1,UD
1.000
design model ω1 = 1.000 ω1,UD
0.366
refined basis, re-held ω1 = 1.000 ω1,UD
0.358
FE model, as designed ω1 = 0.986 ω1,UD
0.364
FE model, re-matched ω1 = 1.000 ω1,UD
0.352
full plate (SSM), as designed ω1 = 1.008 ω1,UD
0.359
full plate (SSM), re-matched ω1 = 1.000 ω1,UD
0.367
full plate (EPMC), re-matched ω1 = 1.000 ω1,UD
0.370
pointwise ceiling (unmanufacturable; estimate)
0.335
J/J_UD at held frequency